
Table of Contents

.....	1
Load Talley's mixed spark + noise test file	1
Magnitude	1
Smooth the magnitude	1
Estimate baseline and threshold	1
Find candidate peaks	2
Convert sample locations to time	2
Print detected events	2
Time difference between events	5
Plot first 100 ms	7
Plot threshold	8
Only show detected events inside first 100 ms	9
Simple summary	10

```
clear;
clc;
close all;
```

Load Talley's mixed spark + noise test file

```
data = load('SPARK_NOISE_EQUAL_RMS_001.mat');

iq = data.eventIQ;
fs = data.fs;
fc = data.fc;

fprintf('Sample rate: %.0f Hz\n', fs);
fprintf('Center frequency: %.1f MHz\n\n', fc/1e6);

Sample rate: 2400000 Hz
Center frequency: 145.9 MHz
```

Magnitude

```
mag = abs(iq);
```

Smooth the magnitude

```
windowLength = 200;
smoothMag = movmean(mag, windowLength);
```

Estimate baseline and threshold

```
baseline = median(smoothMag);
```

```
% Simple threshold above the background level
threshold = baseline * 1.5;

fprintf('Baseline level: %.6f\n', baseline);
fprintf('Detection threshold: %.6f\n\n', threshold);

Baseline level: 0.026469
Detection threshold: 0.039703
```

Find candidate peaks

Minimum spacing of 2 ms between detected events

```
minPeakDistance = round(0.002 * fs);

[peakValues, peakLocations] = findpeaks( ...
    smoothMag, ...
    'MinPeakHeight', threshold, ...
    'MinPeakDistance', minPeakDistance);
```

Convert sample locations to time

```
peakTimes = peakLocations / fs;
```

Print detected events

```
fprintf('==== DETECTED EVENTS =====\n');
fprintf('Number of candidate events: %d\n\n', length(peakTimes));

for k = 1:length(peakTimes)
    fprintf('Event %d: %.6f seconds\n', k, peakTimes(k));
end
```

```
==== DETECTED EVENTS =====
Number of candidate events: 125
```

```
Event 1: 0.002928 seconds
Event 2: 0.011260 seconds
Event 3: 0.019654 seconds
Event 4: 0.027928 seconds
Event 5: 0.036260 seconds
Event 6: 0.044587 seconds
Event 7: 0.052996 seconds
Event 8: 0.056423 seconds
Event 9: 0.061323 seconds
Event 10: 0.069590 seconds
Event 11: 0.077920 seconds
Event 12: 0.086264 seconds
Event 13: 0.094641 seconds
Event 14: 0.102927 seconds
Event 15: 0.111262 seconds
```

Event 16: 0.119595 seconds
Event 17: 0.127934 seconds
Event 18: 0.136192 seconds
Event 19: 0.144598 seconds
Event 20: 0.147979 seconds
Event 21: 0.152977 seconds
Event 22: 0.161259 seconds
Event 23: 0.169645 seconds
Event 24: 0.177863 seconds
Event 25: 0.186263 seconds
Event 26: 0.194658 seconds
Event 27: 0.202925 seconds
Event 28: 0.211315 seconds
Event 29: 0.219530 seconds
Event 30: 0.227858 seconds
Event 31: 0.236256 seconds
Event 32: 0.244655 seconds
Event 33: 0.252975 seconds
Event 34: 0.261194 seconds
Event 35: 0.269590 seconds
Event 36: 0.277969 seconds
Event 37: 0.286264 seconds
Event 38: 0.294527 seconds
Event 39: 0.302862 seconds
Event 40: 0.311193 seconds
Event 41: 0.319660 seconds
Event 42: 0.327990 seconds
Event 43: 0.336258 seconds
Event 44: 0.344594 seconds
Event 45: 0.352935 seconds
Event 46: 0.356414 seconds
Event 47: 0.361303 seconds
Event 48: 0.369454 seconds
Event 49: 0.377918 seconds
Event 50: 0.386259 seconds
Event 51: 0.394595 seconds
Event 52: 0.402972 seconds
Event 53: 0.411197 seconds
Event 54: 0.419532 seconds
Event 55: 0.427922 seconds
Event 56: 0.436307 seconds
Event 57: 0.444652 seconds
Event 58: 0.452979 seconds
Event 59: 0.461261 seconds
Event 60: 0.469653 seconds
Event 61: 0.477918 seconds
Event 62: 0.486267 seconds
Event 63: 0.494660 seconds
Event 64: 0.502931 seconds
Event 65: 0.511199 seconds
Event 66: 0.519593 seconds
Event 67: 0.527971 seconds
Event 68: 0.536197 seconds
Event 69: 0.544591 seconds

Event 70: 0.552987 seconds
Event 71: 0.561325 seconds
Event 72: 0.569530 seconds
Event 73: 0.577928 seconds
Event 74: 0.586121 seconds
Event 75: 0.594527 seconds
Event 76: 0.602923 seconds
Event 77: 0.611318 seconds
Event 78: 0.619525 seconds
Event 79: 0.627923 seconds
Event 80: 0.636329 seconds
Event 81: 0.644599 seconds
Event 82: 0.652863 seconds
Event 83: 0.661266 seconds
Event 84: 0.669532 seconds
Event 85: 0.677979 seconds
Event 86: 0.686313 seconds
Event 87: 0.694585 seconds
Event 88: 0.702972 seconds
Event 89: 0.711194 seconds
Event 90: 0.719595 seconds
Event 91: 0.727860 seconds
Event 92: 0.736318 seconds
Event 93: 0.744657 seconds
Event 94: 0.752861 seconds
Event 95: 0.761262 seconds
Event 96: 0.769638 seconds
Event 97: 0.777985 seconds
Event 98: 0.786197 seconds
Event 99: 0.794661 seconds
Event 100: 0.802866 seconds
Event 101: 0.811199 seconds
Event 102: 0.819530 seconds
Event 103: 0.827927 seconds
Event 104: 0.836193 seconds
Event 105: 0.844638 seconds
Event 106: 0.852918 seconds
Event 107: 0.861263 seconds
Event 108: 0.864895 seconds
Event 109: 0.869599 seconds
Event 110: 0.877866 seconds
Event 111: 0.886261 seconds
Event 112: 0.894454 seconds
Event 113: 0.902931 seconds
Event 114: 0.911321 seconds
Event 115: 0.919530 seconds
Event 116: 0.927924 seconds
Event 117: 0.936256 seconds
Event 118: 0.944526 seconds
Event 119: 0.952990 seconds
Event 120: 0.956753 seconds
Event 121: 0.961261 seconds
Event 122: 0.969654 seconds
Event 123: 0.977920 seconds

Event 124: 0.986198 seconds
Event 125: 0.994527 seconds

Time difference between events

```
if length(peakTimes) > 1

    eventSpacing = diff(peakTimes);

    fprintf('\n===== EVENT SPACING =====\n');

    for k = 1:length(eventSpacing)
        fprintf('Spacing %d: %.6f seconds\n', ...
            k, eventSpacing(k));
    end

    fprintf('\nAverage spacing: %.6f seconds\n', ...
        mean(eventSpacing));

    fprintf('Average spacing: %.3f ms\n', ...
        mean(eventSpacing) * 1000);
end
```

```
===== EVENT SPACING =====
Spacing 1: 0.008332 seconds
Spacing 2: 0.008393 seconds
Spacing 3: 0.008274 seconds
Spacing 4: 0.008333 seconds
Spacing 5: 0.008326 seconds
Spacing 6: 0.008409 seconds
Spacing 7: 0.003427 seconds
Spacing 8: 0.004899 seconds
Spacing 9: 0.008267 seconds
Spacing 10: 0.008331 seconds
Spacing 11: 0.008343 seconds
Spacing 12: 0.008377 seconds
Spacing 13: 0.008286 seconds
Spacing 14: 0.008335 seconds
Spacing 15: 0.008334 seconds
Spacing 16: 0.008338 seconds
Spacing 17: 0.008259 seconds
Spacing 18: 0.008405 seconds
Spacing 19: 0.003381 seconds
Spacing 20: 0.004999 seconds
Spacing 21: 0.008281 seconds
Spacing 22: 0.008386 seconds
Spacing 23: 0.008218 seconds
Spacing 24: 0.008400 seconds
Spacing 25: 0.008395 seconds
Spacing 26: 0.008266 seconds
Spacing 27: 0.008391 seconds
Spacing 28: 0.008215 seconds
```

Spacing 29: 0.008328 seconds
Spacing 30: 0.008398 seconds
Spacing 31: 0.008399 seconds
Spacing 32: 0.008320 seconds
Spacing 33: 0.008219 seconds
Spacing 34: 0.008396 seconds
Spacing 35: 0.008380 seconds
Spacing 36: 0.008295 seconds
Spacing 37: 0.008263 seconds
Spacing 38: 0.008336 seconds
Spacing 39: 0.008331 seconds
Spacing 40: 0.008466 seconds
Spacing 41: 0.008330 seconds
Spacing 42: 0.008268 seconds
Spacing 43: 0.008336 seconds
Spacing 44: 0.008342 seconds
Spacing 45: 0.003478 seconds
Spacing 46: 0.004890 seconds
Spacing 47: 0.008150 seconds
Spacing 48: 0.008465 seconds
Spacing 49: 0.008340 seconds
Spacing 50: 0.008337 seconds
Spacing 51: 0.008376 seconds
Spacing 52: 0.008225 seconds
Spacing 53: 0.008335 seconds
Spacing 54: 0.008390 seconds
Spacing 55: 0.008385 seconds
Spacing 56: 0.008345 seconds
Spacing 57: 0.008327 seconds
Spacing 58: 0.008283 seconds
Spacing 59: 0.008392 seconds
Spacing 60: 0.008265 seconds
Spacing 61: 0.008349 seconds
Spacing 62: 0.008392 seconds
Spacing 63: 0.008272 seconds
Spacing 64: 0.008268 seconds
Spacing 65: 0.008393 seconds
Spacing 66: 0.008378 seconds
Spacing 67: 0.008226 seconds
Spacing 68: 0.008395 seconds
Spacing 69: 0.008396 seconds
Spacing 70: 0.008337 seconds
Spacing 71: 0.008205 seconds
Spacing 72: 0.008398 seconds
Spacing 73: 0.008193 seconds
Spacing 74: 0.008406 seconds
Spacing 75: 0.008395 seconds
Spacing 76: 0.008395 seconds
Spacing 77: 0.008208 seconds
Spacing 78: 0.008397 seconds
Spacing 79: 0.008407 seconds
Spacing 80: 0.008270 seconds
Spacing 81: 0.008263 seconds
Spacing 82: 0.008404 seconds

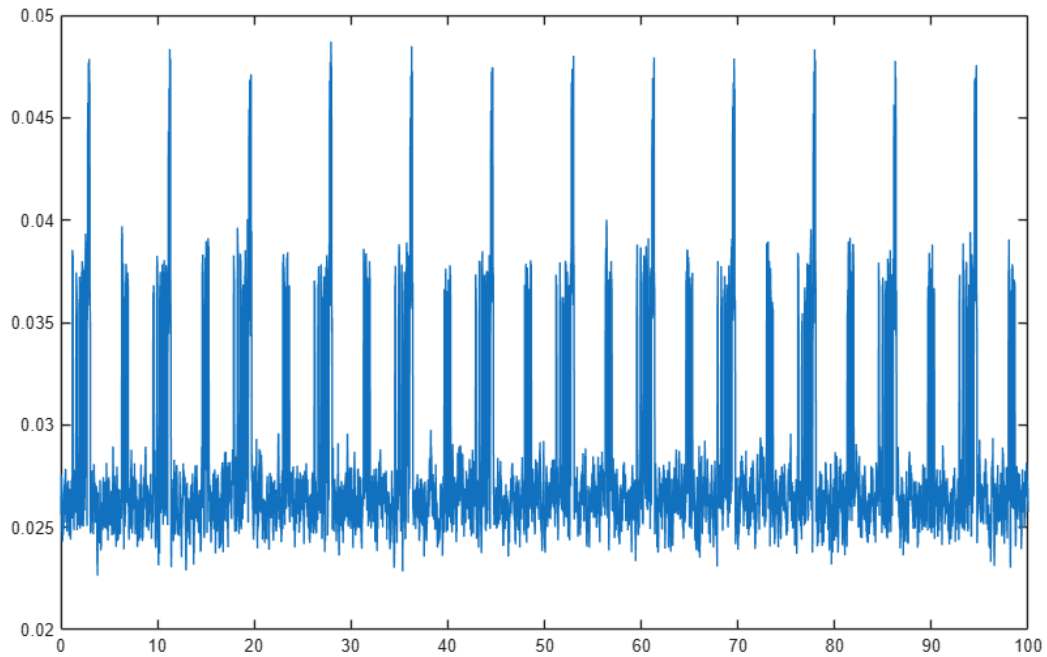
Spacing 83: 0.008265 seconds
Spacing 84: 0.008447 seconds
Spacing 85: 0.008335 seconds
Spacing 86: 0.008272 seconds
Spacing 87: 0.008387 seconds
Spacing 88: 0.008222 seconds
Spacing 89: 0.008401 seconds
Spacing 90: 0.008265 seconds
Spacing 91: 0.008458 seconds
Spacing 92: 0.008340 seconds
Spacing 93: 0.008204 seconds
Spacing 94: 0.008401 seconds
Spacing 95: 0.008376 seconds
Spacing 96: 0.008346 seconds
Spacing 97: 0.008212 seconds
Spacing 98: 0.008464 seconds
Spacing 99: 0.008205 seconds
Spacing 100: 0.008333 seconds
Spacing 101: 0.008330 seconds
Spacing 102: 0.008397 seconds
Spacing 103: 0.008267 seconds
Spacing 104: 0.008445 seconds
Spacing 105: 0.008280 seconds
Spacing 106: 0.008345 seconds
Spacing 107: 0.003632 seconds
Spacing 108: 0.004705 seconds
Spacing 109: 0.008267 seconds
Spacing 110: 0.008395 seconds
Spacing 111: 0.008193 seconds
Spacing 112: 0.008477 seconds
Spacing 113: 0.008390 seconds
Spacing 114: 0.008208 seconds
Spacing 115: 0.008395 seconds
Spacing 116: 0.008332 seconds
Spacing 117: 0.008270 seconds
Spacing 118: 0.008463 seconds
Spacing 119: 0.003763 seconds
Spacing 120: 0.004508 seconds
Spacing 121: 0.008393 seconds
Spacing 122: 0.008266 seconds
Spacing 123: 0.008278 seconds
Spacing 124: 0.008329 seconds

Average spacing: 0.007997 seconds
Average spacing: 7.997 ms

Plot first 100 ms

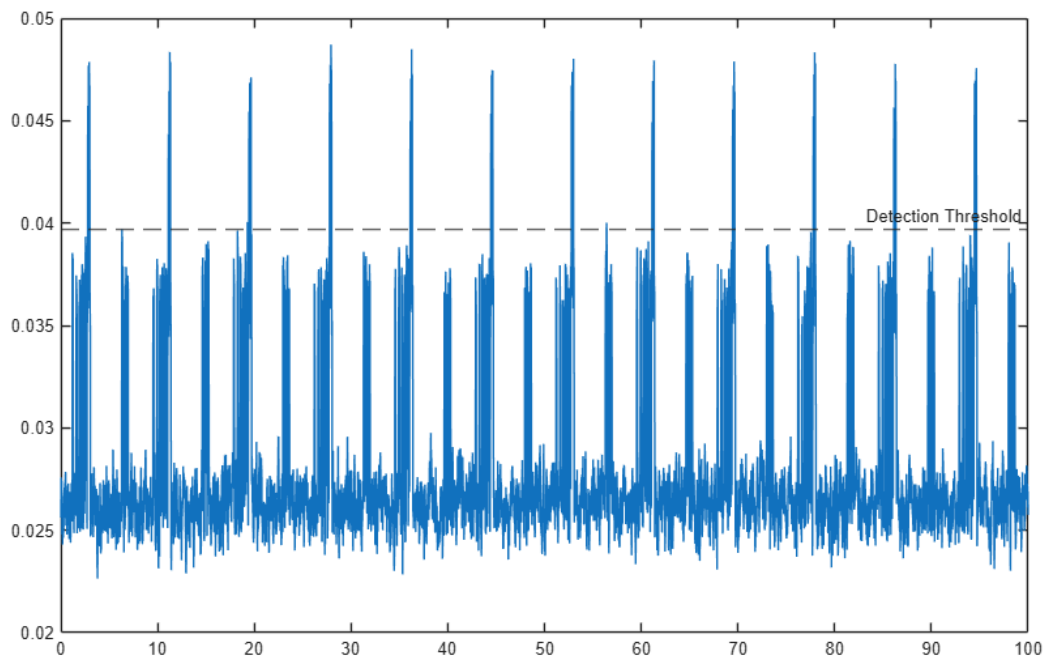
```
showTime = 0.1;  
Nshow = round(showTime * fs);  
  
t = (0:length(iq)-1) / fs;
```

```
figure;  
  
plot(t(1:Nshow)*1000, smoothMag(1:Nshow));  
hold on;
```



Plot threshold

```
ylines(threshold, '--', 'Detection Threshold');
```



Only show detected events inside first 100 ms

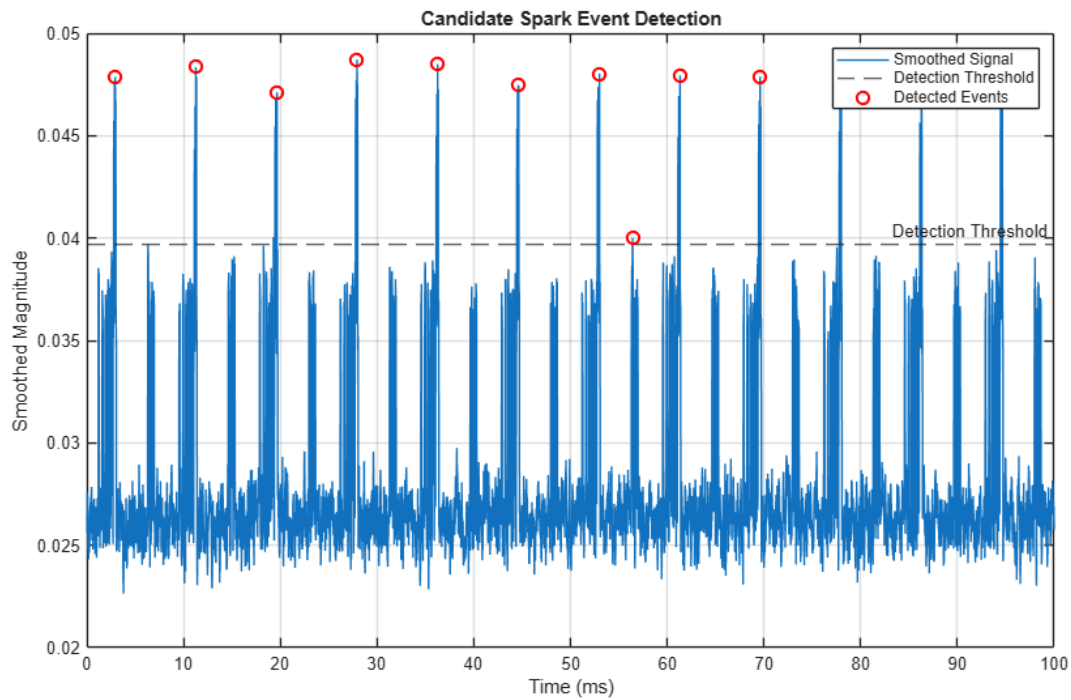
```
visibleEvents = peakLocations <= Nshow;

plot( ...
    peakTimes(visibleEvents)*1000, ...
    peakValues(visibleEvents), ...
    'ro', ...
    'MarkerSize', 7, ...
    'LineWidth', 1.5);

xlabel('Time (ms)');
ylabel('Smoothed Magnitude');
title('Candidate Spark Event Detection');
grid on;

legend( ...
    'Smoothed Signal', ...
    'Detection Threshold', ...
    'Detected Events');

hold off;
```



Simple summary

```
fprintf('\n===== SUMMARY =====\n');  
  
if isempty(peakTimes)  
    fprintf('NO CANDIDATE EVENTS DETECTED\n');  
else  
    fprintf('CANDIDATE SPARK EVENTS DETECTED\n');  
end
```

```
===== SUMMARY =====  
CANDIDATE SPARK EVENTS DETECTED
```

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